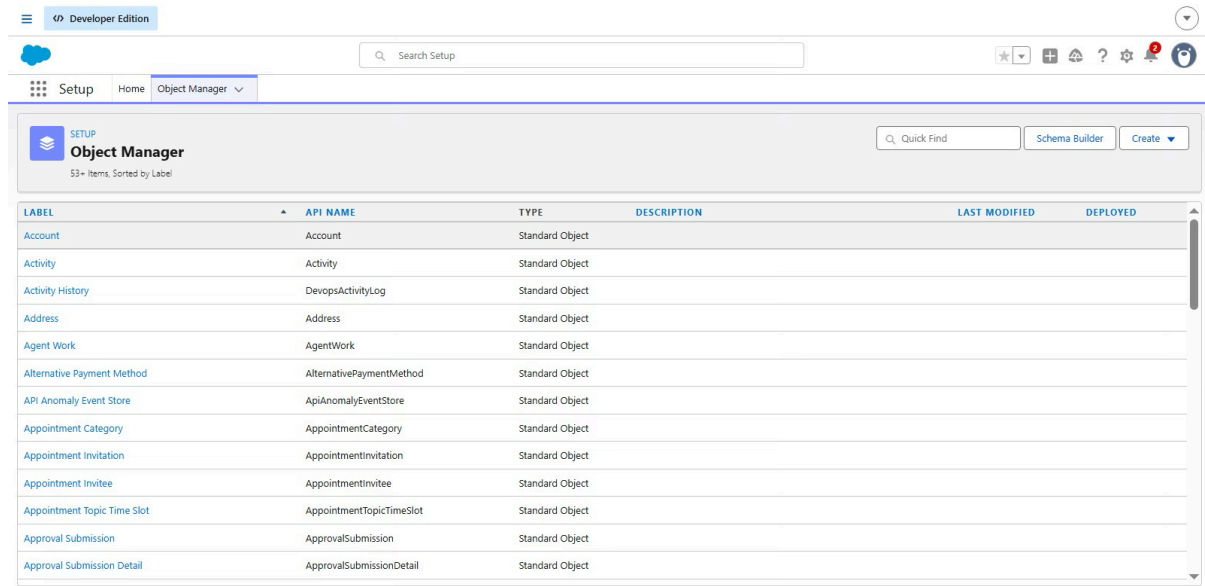


Chapter 2 - Requirement Analysis

2.1 Requirement Analysis

Requirement Analysis is the initial phase of the project lifecycle where business needs are identified and translated into technical requirements. For the AI-Powered Customer Support & Service Automation project, this phase focused on defining the scope, objectives, automation requirements, AI capabilities, security requirements, and implementation strategy before development began.

The analysis identified the need for centralized customer support management using Salesforce. Business processes such as customer management, product management, order tracking, and support case handling were analyzed. Salesforce features including Custom Objects, Validation Rules, Record-Triggered Flows, Queues, Lightning App Builder, Prompt Builder, and Agentforce were selected to satisfy these requirements.



The screenshot displays the Salesforce Object Manager interface. At the top, there's a navigation bar with 'Developer Edition' and a search bar labeled 'Search Setup'. Below this, the 'Object Manager' tab is selected. The main area shows a list of 53 items, sorted by label. The table below represents the data shown in this list.

LABEL	API NAME	TYPE	DESCRIPTION	LAST MODIFIED	DEPLOYED
Account	Account	Standard Object			
Activity	Activity	Standard Object			
Activity History	DevopsActivityLog	Standard Object			
Address	Address	Standard Object			
Agent Work	AgentWork	Standard Object			
Alternative Payment Method	AlternativePaymentMethod	Standard Object			
API Anomaly Event Store	ApiAnomalyEventStore	Standard Object			
Appointment Category	AppointmentCategory	Standard Object			
Appointment Invitation	AppointmentInvitation	Standard Object			
Appointment Invitee	AppointmentInvitee	Standard Object			
Appointment Topic Time Slot	AppointmentTopicTimeSlot	Standard Object			
Approval Submission	ApprovalSubmission	Standard Object			
Approval Submission Detail	ApprovalSubmissionDetail	Standard Object			

2.2 Business Requirements

The project aims to provide a centralized customer support platform capable of storing customer, product, order, and support case information. Business requirements include automated case routing, AI-assisted case summary generation, secure data management, improved customer response time, reduced manual effort, and scalable Salesforce-based automation.

2.3 Functional Requirements

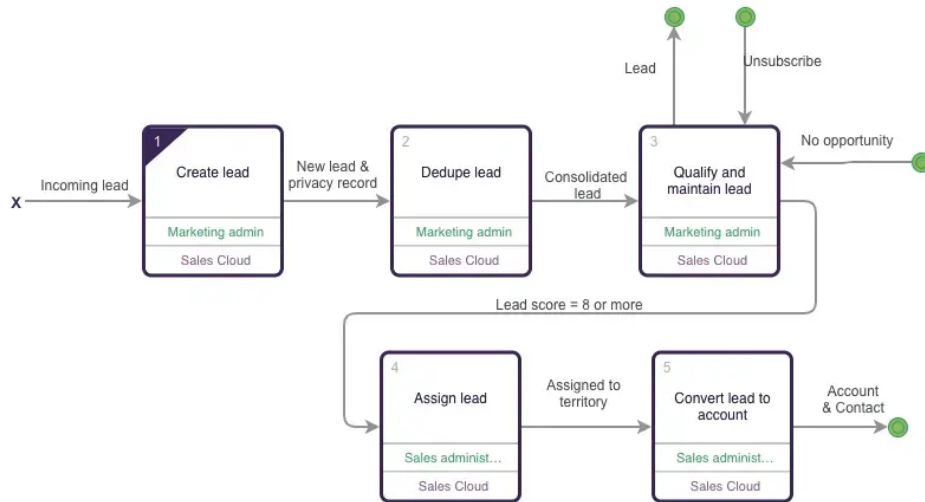
Module	Functional Requirement
Customer Management	Create and manage customer records including contact information.
Product Management	Maintain product information and warranty details.
Order Management	Create orders and relate them to customers and products.
Use Case Management	Create support cases with priority and issue category.
Automation	Automatically assign cases using Flows and Queues.
Artificial Intelligence	Generate AI summaries using Prompt Builder.

2.4 Non-Functional Requirements

- Performance: Fast execution of flows and record operations.
- Security: Profile-based access, field-level security, and validation rules.
- Reliability: Consistent automation and accurate record management.
- Scalability: Ability to support additional users, AI capabilities, and integrations.
- Usability: User-friendly Lightning interface with simple navigation.
- Availability: Cloud-based Salesforce platform ensuring high availability.

2.5 Stakeholder Analysis

Stakeholder	Responsibility
Project Sponsor	Defines objectives and approves deliverables.
Salesforce Administrator	Configures objects, flows, Prompt Builder, and security.
Customer Support Team	Creates and manages support cases.
Management	Reviews reports and monitors performance.
IT Support	Maintains the Salesforce environment.
Customers	Receive improved customer support services.



2.6 Scope of the Project

The project focuses on implementing an AI-enabled customer support solution using Salesforce. It includes custom objects, lookup relationships, validation rules, record-triggered flows, queues, Lightning pages, and Prompt Builder integration to automate support activities while maintaining secure and accurate data.

2.7 Assumptions

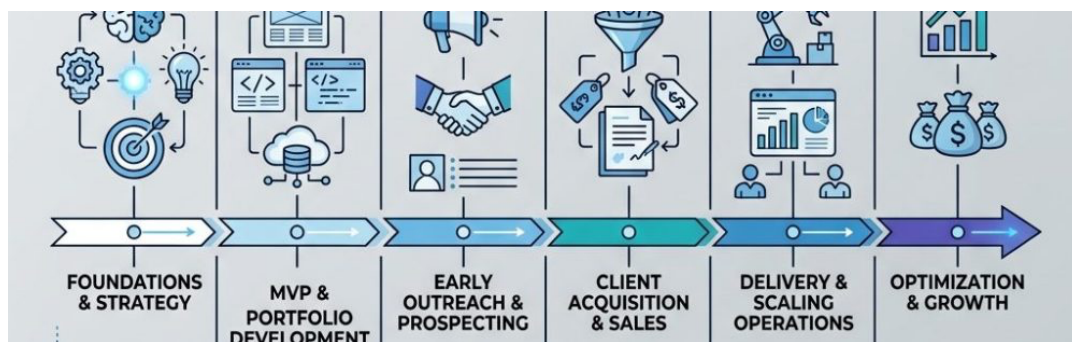
- Salesforce Developer Edition is available.
- Prompt Builder and AI features are enabled.
- Users have required permissions.
- Sample records are available for testing.

2.8 Constraints

- Limited to Salesforce Developer Edition features.
- Uses sample business data.
- No third-party integrations implemented.

2.9 Execution Roadmap

Phase	Description
Phase 1	Requirement Analysis and Planning
Phase 2	Salesforce Configuration
Phase 3	Lightning UI Development
Phase 4	AI Integration using Prompt Builder
Phase 5	Testing and Deployment



2.10 Chapter Summary

The requirement analysis phase established the business objectives, identified system functionality, selected Salesforce technologies, and defined the overall implementation roadmap. This phase provided the foundation for developing a secure, scalable, and AI-enabled customer support solution.